COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 2 : SLO – 2 | BCD, Gray code, Excess-3
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Multiple Choice Questions

• 1. c — 0110 (add 6 correction)
• 2. c — Copy the first bit
• 3. c — 1011 (decimal 11 is not a valid BCD digit)
• 4. c — Better for analog-to-digital conversion (only one bit changes between consecutive values)
• 5. b — Handling sums greater than 9 in BCD

Two-Mark Questions

• 1. BCD (Binary-Coded Decimal) represents each decimal digit 0–9 with its own 4-bit binary code (8421 weighted). Application: digital clocks/calculators and financial systems where exact decimal display is needed.
• 2. Gray to binary (5-bit): (i) Copy the MSB of Gray as MSB of binary. (ii) Each next binary bit = previous BINARY bit XOR current Gray bit. (iii) Repeat left to right for all 5 bits.

Scenario-Based Long Answer

• Correction rule: after adding two BCD digits, if the 4-bit result is greater than 1001 (9) or a carry is generated, add 0110 (6) to that digit and propagate the carry to the next higher digit.
• Example: 9 + 3: 1001 + 0011 = 1100 (12, invalid BCD). Add 0110 → 1100 + 0110 = 1 0010. Result: 0010 in the units digit and a carry of 1 to the tens digit, i.e. 0001 0010 = 12 in BCD. The clock's display logic applies this per digit so every digit stays a valid 0–9 code.

Match the Following

• BCD — a) 8,4,2,1 weighted code
• Excess-3 — b) Non-weighted, self-complementary code
• Gray Code — c) Used for minimizing errors in A/D conversion
• Invalid BCD Code — d) 1011
• Binary to Gray Conversion — e) Copy first bit, then XOR

Fill in the Blanks

• 1. 0111 2. 0110 3. MSB (first/most significant) 4. 0110 5. errors

Scenario-Based Small Activity

• Convert binary 1010 to Gray: (i) Copy first bit: g3 = 1. (ii) g2 = b3 XOR b2 = 1 XOR 0 = 1. (iii) g1 = b2 XOR b1 = 0 XOR 1 = 1. (iv) g0 = b1 XOR b0 = 1 XOR 0 = 1.
• Output Gray code = 1111. Because only one bit changes between consecutive Gray values, transmitting 1111 avoids multi-bit transition errors from the sensor.

True or False

• 1. FALSE — Only one bit changes between consecutive Gray values. 2. FALSE — BCD is not efficient for high-speed arithmetic (needs corrections).
• 3. TRUE 4. FALSE — Bits are XORed, not reversed. 5. TRUE